4.5 The dimension of a vector space

In this section $\mathcal{B} = \{b_1, \dots, b_n\}$ is a basis for the (real) vector space V.

Def. $\dim V = n$

Thm 11
Every basis of V has n elements

Thm 10
If $\{v_1, \dots, v_p\} \subseteq V$ and $p > \dim V = n$ then $\{v_1, \dots, v_p\}$ is lin. dependent.

Thm 13
① Any lin. ind. $\{v_1, \dots, v_n\} \subseteq V$ is a basis for V

② Any $\{v_1, \dots, v_n\} \subseteq V$ with $\operatorname{Span}\{v_1, \dots, v_n\} = V$ is a basis for V.

Example
Recall that $\mathcal{B} = \{1, x, x^2\}$ is a basis for $\mathbb{P}_2$.
$\dim \mathbb{P}_2 = 3$ since 3 vectors in $\mathcal{B}$.
Is $\mathcal{W} = \{1, 1+x, 1+x+x^2\}$ a basis for $\mathbb{P}_2$?
Let's check lin. independence.

$$c_1(1) + c_2(1+x) + c_3(1+x+x^2) = 0 \quad (*)$$

$\forall x \in \mathbb{R}$.

We hope that $c_1 = c_2 = c_3 = 0$ is the <u>only</u> solution.

We convert (*) into

$$(c_1 + c_2 + c_3)(1) + (c_2 + c_3)x + c_3 x^2 = 0$$

or
$$\begin{cases} c_1 + c_2 + c_3 = 0 \\ c_2 + c_3 = 0 \\ c_3 = 0 \end{cases}$$

Clearly $c_1 = c_2 = c_3$ is the only solution to this system. Therefore $\mathcal{W}$ is lin. ind.

Thm 13 ① tells us that $\mathcal{W} = \{1, 1+x, 1+x+x^2\}$ is a basis, since it is lin. ind. and has $3 = \dim \mathbb{P}_2$ vectors in it.

Define
$$p_1(x) = 1$$
$$p_2(x) = 1 + x$$
$$p_3(x) = 1 + x + x^2.$$

$\{\underbrace{p_1, p_2, p_3}_{\text{functions}}\}$ is a basis for $\mathcal{W}$.

More generally $\dim \mathbb{P}_n = n+1$ and any lin. ind $\{p_1, \dots, p_{n+1}\} \subseteq \mathbb{P}_n$ is a basis by Thm 13 ①.

Example

Suppose $\operatorname{span}\{v_1, \dots, v_k\} = V$

If $\{v_1, \dots, v_k\}$ is lin. ind. then $\{v_1, \dots, v_k\}$ is a basis and $k = n = \dim V$ by Thm 11.

Otherwise $\{v_1, \dots, v_k\}$ is lin. dependent. and some of the vectors in this set are linear combinations of the others. (Thm 4 §4.3) We may throw out these "bad" vectors and then are left with a smaller set

$$\{v_{j_1}, \dots, v_{j_\ell}\} \subseteq \{v_1, \dots, v_k\}$$

such that $\ell < k$, $\operatorname{Span}\{v_{j_1}, \dots, v_{j_\ell}\} = V$ (Thm 5 §4.3), and $\{v_{j_1}, \dots, v_{j_\ell}\}$ is lin. independent. In particular $\{v_{j_1}, \dots, v_{j_\ell}\}$ is a basis for V.

Theorem

Every spanning set $\{v_1, \dots, v_k\}$ for V contains a basis and $\dim V = n \leq k$.

Example

Suppose $\{v_1, \dots, v_k\} \subseteq V$ is lin. ind.

If $\operatorname{Span}\{v_1, \dots, v_k\} = V$ the $\{v_1, \dots, v_k\}$ is a basis and $k = n = \dim V$.

Otherwise $\operatorname{Span}\{v_1, \dots, v_k\} \subsetneq V$.

Pick $v_{k+1} \in V$ with $v_{k+1} \notin \operatorname{Span}\{v_1, \dots, v_k\}$.

This implies $v_{k+1} \neq d_1 v_1 + \dots + d_k v_k$ for <u>any</u> $d_1, \dots, d_k \in \mathbb{R}$. Moreover, Thm 4 tells us that $v_j \neq d_1 v_1 + \dots + d_{j-1} v_{j-1}$ for <u>any</u> $d_1, \dots, d_j \in \mathbb{R}$ and $j \leq k$, as $\{v_1, \dots, v_k\}$ is lin. ind.

By Thm 4 the bigger set $\{v_1, \dots, v_k, v_{k+1}\}$ is lin. ind. By construction $\operatorname{span}\{v_1, \dots, v_{k+1}\}$ is bigger than $\operatorname{span}\{v_1, \dots, v_k\}$.

If $\{v_1, \dots, v_{k+1}\}$ spans V then it is a basis. and $k+1 = n = \dim V$.

Otherwise we repeat the above procedure to obtain a lin. ind set $\{v_1, \dots, v_{k+1}, v_{k+2}\}$ with a bigger span. This procedure terminates by Thm 10.

Theorem.

Suppose $\{v_1, \dots, v_k\} \subseteq V$ is lin. ind.

Then $\{v_1, \ldots, v_k\}$ can be extend to a basis $\{v_1, \ldots, v_k, v_{k+1}, \ldots, v_n\}$, and $k \leq \dim V = n$.

Theorem 12

Suppose $H \subseteq V$ is a subspace of V. Then any lin. ind. subset of H can be extended to a basis for H, and $\dim H \leq \dim V$

(Moreover $\dim H = \dim V \iff H = V$).

Closing thoughts

- Main thing about a basis $\mathcal{B} = \{b_1, \ldots, b_n\}$ for V is that every $v \in V$ can be written as a linear combination

$$v = c_1 b_1 + \cdots + c_n b_n$$

for a unique $c_1, \ldots, c_n \in \mathbb{R}$.

- Knowing $\dim V$ gives us information about lin. ind. and spanning. It also helps in identifying bases. (Thm 13)

- We have assumed that V has a <u>finite</u> basis. $\mathbb{P}$, for example, does <u>not</u> have a finite basis. ($\mathbb{P}$ has basis $\{1, x, x^2, x^3, \ldots\}$.
If a vector space does <u>not</u> have a finite basis we call it <u>infinite-dimensional</u>. The

theorems of this section do not apply to infinite-dimensional vector spaces.

4.6 Change of basis

In this section V is a (real) vector space with basis $\mathcal{B} = \{b_1, \dots, b_n\}$, so $n = \dim V$.

Recall that $[v]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n$

where $v = c_1 b_1 + \dots + c_n b_n$.

Now suppose $\mathcal{W} = \{w_1, \dots, w_n\}$ is another basis for V.

Then $v = d_1 w_1 + \dots + d_n w_n$ for unique $d_1, \dots, d_n \in \mathbb{R}$

and $[v]_{\mathcal{W}} = \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix} \in \mathbb{R}^n$.

$$\begin{tikzcd}
 & V \arrow[dl, leftrightarrow, "{[\cdot]_{\mathcal{W}}}"'] \arrow[dr, leftrightarrow, "{[\cdot]_{\mathcal{B}}}"] & \\
\mathbb{R}^n & & \mathbb{R}^n \arrow[ll, dashed, "?"]
\end{tikzcd}$$

Thm 9
$[\cdot]_{\mathcal{B}}$ is a lin. trans. and is invertible (1-1 and onto)

The map in question is linear and given by matrix mult.
What is the matrix.

Example 1 in text.

Suppose $n=2$ above and $v = 3b_1 + (1)b_2 \in V$

Then $[v]_B = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \in \mathbb{R}^2$.

How can we find $[v]_{\mathcal{W}}$ for $\mathcal{W} = \{w_1, w_2\}$?

For this we need to write b_1 and b_2 in terms of $\mathcal{W}$. Let's just suppose $b_1 = 4w_1 + (1)w_2$

$b_2 = -6w_1 + w_2$. Then

$$[b_1]_{\mathcal{W}} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \text{ and } [b_2]_{\mathcal{W}} = \begin{bmatrix} -6 \\ 1 \end{bmatrix}.$$

It follows that

$$\begin{aligned}
[v]_{\mathcal{W}} &= [3b_1 + (1)b_2]_{\mathcal{W}} \\
&= 3[b_1]_{\mathcal{W}} + (1)[b_2]_{\mathcal{W}} && ([\cdot]_{\mathcal{W}} \text{ is linear}). \\
&= 3\begin{bmatrix} 4 \\ 1 \end{bmatrix} + (1)\begin{bmatrix} -6 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 4 & -6 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 3 \\ 1 \end{bmatrix} && (\text{matrix mult.}) \\
&= \begin{bmatrix} 4 & -6 \\ 1 & 1 \end{bmatrix}[v]_B \\
&= \left[\, [b_1]_{\mathcal{W}} \;\; [b_2]_{\mathcal{W}} \,\right][v]_B.
\end{aligned}$$

Theorem 15

Let $B = \{b_1, \ldots, b_n\}$ and $\mathcal{W} = \{w_1, \ldots, w_n\}$ be bases for V. Then

$$[v]_{\mathcal{W}} = \underset{\mathcal{W} \leftarrow B}{P}\, [v]_B$$

where $\underset{\mathcal{W} \leftarrow \mathcal{B}}{P} \in M_{n \times n}$ is defined by

$$\underset{\mathcal{W} \leftarrow \mathcal{B}}{P} = \left[[b_1]_{\mathcal{W}} \cdots [b_n]_{\mathcal{W}} \right].$$

Def $\underset{\mathcal{W} \leftarrow \mathcal{B}}{P}$ is called the <u>change of basis matrix</u> from $\mathcal{B}$ to $\mathcal{W}$.